Negating Quantified Statements

The negation of a statement changes it from true to false, or false to true.

If P is a proposition, then $\neg(P)$ is its negation

e.g., $\exists x \in \mathbb{Q} : x^2 = 2$, this is false

negation $\to$ $\forall x \in \mathbb{Q} : x^2 \neq 2$, which is true

When negating a statement, you flip the universal quantifiers, alongside the truth or falsities.

$\forall \leftrightarrow \exists$ $\quad$ $< \leftrightarrow \geq$

$= \leftrightarrow \neq$

Implication

The implication of one statement because of another is to say that the implied statement follows logically from the previous.

P	Q	$P \Rightarrow Q$
F	F	T
F	T	T
T	F	F
T	T	T

Though it seems illogical, it is very much so true. In everyday speech, yes it would be different.

Since the statements should follow each other logically, then this must be false

General methods of proofs

- Direct Proofs

simply take you from point A to point B

these are good for statements of the form $\forall x \in S : P(x) \Rightarrow Q(x)$.

Let $x \in S$, then assume $P(x)$ is true, and finally deduce that $Q(x)$ is true

- Contrapositive Proof

Essentially, consider the logically equivalent opposite to the original statement and perform a direct proof from there.

$P \Rightarrow Q$ becomes $\neg(Q) \Rightarrow \neg(P)$

This is useful for when the contrapositive of a statement is easier to prove than the original statement.

e.g., $x \notin \mathbb{Q} \Rightarrow \frac{1}{x} \notin \mathbb{Q}$

The contrapositive is $\frac{1}{x} \in \mathbb{Q} \Rightarrow x \in \mathbb{Q}$, which is far easier to prove

- Proof by Contradiction

This is to assume the negation and deduce a contradiction, implying that that the original statement is true.

Take the statement P and assume negation which is $\neg(P)$.
You could also say $\neg(P) \Rightarrow F$ from which we can deduce P

These proofs tend to rely more heavily on definitions